

## Operators in Python

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An operator is used to perform specific mathematical or logical operation on values. The values that the operator work on are called operands.

For example  $10 + \text{num}$ , The value '10' and the variable 'num' are operands and the  $+$  (plus) sign is an operator. Python supports several kind of operators whose categorization is briefly explained:

**Arithmetic Operators:** Python support arithmetic operators that are used to perform. The basic arithmetic operations, as well as modular division, floor, division and exponentiation.

| Operator | Operation      | Description                                                                                                               | Examples                                                                                                                                                                                                                                          |
|----------|----------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| +        | Addition       | Adds the two numeric values on either side of the operator.                                                               | <pre>n1 = 9 n2 = 7 sum = n1 + n2 sum 16 print(sum) 16 print("sum =",sum) sum = 16</pre> <p><i>Handwritten notes: 'operator' points to '+', 'operand' points to 'n1' and 'n2'.</i></p>                                                             |
|          |                | This operator can also be used to concatenate two strings on either side of the operator.                                 | <pre>fname = 'Vaanya' lname = 'Sharma' fullName = fname + ' ' + lname print("Full Name:",fullName) Full Name: Vaanya Sharma</pre> <p><i>Handwritten notes: 'operator' points to '+', 'space' points to the space character in the string.</i></p> |
| -        | Subtraction    | Subtracts the operand on the right from the operand on the left.                                                          | <pre>n1 9 n2 7 diff = n1 - n2 diff 2</pre>                                                                                                                                                                                                        |
| *        | Multiplication | Multiply is the two values on both side of the operator.                                                                  | <pre>n1 9 n2 7 prod = n1 * n2 print('Product =', prod) Product = 63</pre>                                                                                                                                                                         |
|          |                | Repeats the item on left of the operator if first operand is a string and 2nd operand is an integer value. Or vice versa. | <pre>fname 'Vaanya' R = fname * 5 print(R) VaanyaVaanyaVaanyaVaanyaVaanya R = 5 * fname print(R) VaanyaVaanyaVaanyaVaanyaVaanya R = 10 * fname</pre>                                                                                              |

|  |    |                |                                                                                                                                                                 |                                                                  |
|--|----|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
|  |    |                |                                                                                                                                                                 | print(R)                                                         |
|  |    |                |                                                                                                                                                                 | VaanyaVaanyaVaanyaVaanyaVaanyaVaanya<br>VaanyaVaanyaVaanyaVaanya |
|  |    |                |                                                                                                                                                                 | line = '_' * 20                                                  |
|  |    |                |                                                                                                                                                                 | print(line)                                                      |
|  |    |                |                                                                                                                                                                 | _____                                                            |
|  | /  | Division       | Divides the operand on the left by the operand on the right, and the returns the quotient.                                                                      | n1                                                               |
|  |    |                |                                                                                                                                                                 | 9                                                                |
|  |    |                |                                                                                                                                                                 | n2                                                               |
|  |    |                |                                                                                                                                                                 | 7                                                                |
|  |    |                |                                                                                                                                                                 | d = n1 / n2                                                      |
|  |    |                |                                                                                                                                                                 | print('Result =', d)                                             |
|  |    |                |                                                                                                                                                                 | Result = 1.2857142857142858                                      |
|  |    |                |                                                                                                                                                                 | type(d)                                                          |
|  |    |                |                                                                                                                                                                 | <class 'float'>                                                  |
|  |    |                |                                                                                                                                                                 | type(n1)                                                         |
|  |    |                |                                                                                                                                                                 | <class 'int'>                                                    |
|  |    |                |                                                                                                                                                                 | type(n2)                                                         |
|  |    |                |                                                                                                                                                                 | <class 'int'>                                                    |
|  | %  | Modulus        | Divide the operand on the left by the operand on the right and returns the remainder.                                                                           | n1                                                               |
|  |    |                |                                                                                                                                                                 | 9                                                                |
|  |    |                |                                                                                                                                                                 | n2                                                               |
|  |    |                |                                                                                                                                                                 | 7                                                                |
|  |    |                |                                                                                                                                                                 | rem = n1 % n2                                                    |
|  |    |                |                                                                                                                                                                 | print('Remainder =', rem)                                        |
|  |    |                |                                                                                                                                                                 | Remainder = 2                                                    |
|  | // | Floor division | Divide the operand on the left by the operand on the right and returns the quotient by removing the decimal part. It is sometimes also called integer division. | n1                                                               |
|  |    |                |                                                                                                                                                                 | 9                                                                |
|  |    |                |                                                                                                                                                                 | n2                                                               |
|  |    |                |                                                                                                                                                                 | 7                                                                |
|  |    |                |                                                                                                                                                                 | q = n1 // n2                                                     |
|  |    |                |                                                                                                                                                                 | print('Quotient =', q)                                           |
|  |    |                |                                                                                                                                                                 | Quotient = 1                                                     |
|  | ** | Exponent       | Performs exponential (power) calculation on operands. That is, raise the operand on the left to the power of the operand on the right.                          | a = 2                                                            |
|  |    |                |                                                                                                                                                                 | b = 5                                                            |
|  |    |                |                                                                                                                                                                 | e = a ** 5                                                       |
|  |    |                |                                                                                                                                                                 | print(a,'**', 5,' =', e)                                         |

$$2 ** 5 = 32$$

## Relational operators

Relational operators compare the values of operands on its either side and determines the relationship among them. The result is always a 'bool' value(True or False).

num1 = 10, num2 = 0, num3 = 10, str1 = 'Good', str2 = 'Afternoon'

| Operator | Operation             | Description                                                                                                                                           | Example                                                                                                                  |
|----------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| ==       | Equal to              | If the values of two operands are equal then the condition is true, otherwise it is false.                                                            | num1 == num2<br>False<br>str1 == str2<br>False<br>num1 == 10<br>True                                                     |
| !=       | Not equal to          | If values of two operands are not equal then condition is true, otherwise it is false.                                                                | num1 != num2<br>True<br>str1 != str2<br>True<br>str2 != 'Afternoon'<br>False <i>it means str2 is storing 'Afternoon'</i> |
| >        | Greater than          | If the value of the left side operand is greater than the value of the right side operand, then condition is true, otherwise it is false.             | num1 > num2<br>True<br>num1 > num3<br>False                                                                              |
| <        | Less than             | If the value of the left side operand is less than the value of right side operand then condition is true, otherwise it is false.                     | num1 < num2<br>False<br>num2 < num1<br>True                                                                              |
| >=       | Greater than equal to | If the value of the left side operand is greater than or equal to the value of the right side operand, then condition is true. Otherwise it is false. | num1 >= num2<br>True<br>num1 >= num3<br>True<br>num2 >= num3<br>False                                                    |
| <=       | Less than equal to    | If the value of the left operand is less than or equal to the value of right of rent, then condition is true. Otherwise it is false.                  | num1 <= num2<br>False<br>num1 <= num3<br>True                                                                            |

## Assignment Operator

Assignment operator assigns or changes the value of the variable on its left.

| Operator | Description                                                 | Example   |
|----------|-------------------------------------------------------------|-----------|
| =        | Assigns value from right side operand to left side operand. | num1 = 10 |

|  |    |                                                                                                                                                                         |                    |  |
|--|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--|
|  |    |                                                                                                                                                                         | num2 = 0           |  |
|  |    |                                                                                                                                                                         | num3 = 10          |  |
|  |    |                                                                                                                                                                         | str1 = 'Good'      |  |
|  |    |                                                                                                                                                                         | str2 = 'Afternoon' |  |
|  |    |                                                                                                                                                                         | num1 = 10          |  |
|  | += | It adds the value of right side operand to the left side operand and assigns the result to the left side operand.<br>Note: x += y is same as x = x + y                  | x = 5              |  |
|  |    |                                                                                                                                                                         | y = 7              |  |
|  |    |                                                                                                                                                                         | x = x + y          |  |
|  |    |                                                                                                                                                                         | x                  |  |
|  |    |                                                                                                                                                                         | 12                 |  |
|  |    |                                                                                                                                                                         | y                  |  |
|  |    |                                                                                                                                                                         | 7                  |  |
|  |    |                                                                                                                                                                         | x += y             |  |
|  |    |                                                                                                                                                                         | x                  |  |
|  |    |                                                                                                                                                                         | 19                 |  |
|  |    |                                                                                                                                                                         | alpha1 = "ABC"     |  |
|  |    |                                                                                                                                                                         | alpha2 = "DEF"     |  |
|  |    |                                                                                                                                                                         | alpha1 += alpha2   |  |
|  |    |                                                                                                                                                                         | alpha1             |  |
|  |    |                                                                                                                                                                         | 'ABCDEF'           |  |
|  |    |                                                                                                                                                                         | alpha2             |  |
|  |    |                                                                                                                                                                         | 'DEF'              |  |
|  | -= | It subtracts the value of right side operand from the left side operand and assigns the result to the left side operand.<br>Note: x -= y is same as x = x - y           | y                  |  |
|  |    |                                                                                                                                                                         | 7                  |  |
|  |    |                                                                                                                                                                         | x                  |  |
|  |    |                                                                                                                                                                         | 19                 |  |
|  |    |                                                                                                                                                                         | x -= y             |  |
|  |    |                                                                                                                                                                         | x                  |  |
|  |    |                                                                                                                                                                         | 12                 |  |
|  | *= | It multiplies the value of right side operand with the value of left side operand and assigns the result to the left side operand.<br>Note: x *= y is same as x = x * y | x                  |  |
|  |    |                                                                                                                                                                         | 12                 |  |
|  |    |                                                                                                                                                                         | y                  |  |
|  |    |                                                                                                                                                                         | 7                  |  |
|  |    |                                                                                                                                                                         | x *= y             |  |
|  |    |                                                                                                                                                                         | x                  |  |

|     |                                                                                                                                                                   |         |                   |  |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------|--|
|     |                                                                                                                                                                   |         | 84                |  |
|     |                                                                                                                                                                   |         | a = 'India'       |  |
|     |                                                                                                                                                                   |         | a *= 3            |  |
|     |                                                                                                                                                                   |         | a                 |  |
|     |                                                                                                                                                                   |         | 'IndiaIndiaIndia' |  |
|     |                                                                                                                                                                   |         |                   |  |
| /=  | It divides the value of left side operand by the value of right side operand and assign the result to the left side operand.<br>Note: x /= y is same as x = x / y | x       |                   |  |
|     |                                                                                                                                                                   | 84      |                   |  |
|     |                                                                                                                                                                   | y       |                   |  |
|     |                                                                                                                                                                   | 7       |                   |  |
|     |                                                                                                                                                                   | x /= y  |                   |  |
|     |                                                                                                                                                                   | x       |                   |  |
|     |                                                                                                                                                                   | 12.0    |                   |  |
| %=  | It performs modulus operation using the 2 operands and assigns the result to the left side operand.<br>Note: x %= y is same as x = x % y                          | x       |                   |  |
|     |                                                                                                                                                                   | 12.0    |                   |  |
|     |                                                                                                                                                                   | y       |                   |  |
|     |                                                                                                                                                                   | 7       |                   |  |
|     |                                                                                                                                                                   | x %=y   |                   |  |
|     |                                                                                                                                                                   | x       |                   |  |
|     |                                                                                                                                                                   | 5.0     |                   |  |
| //= | It performs floor division using two operands and assigns the result to the left side operand.<br>Note: x //= y is same as x = x // y                             | x       |                   |  |
|     |                                                                                                                                                                   | 5.0     |                   |  |
|     |                                                                                                                                                                   | y       |                   |  |
|     |                                                                                                                                                                   | 7       |                   |  |
|     |                                                                                                                                                                   | x //= y |                   |  |
|     |                                                                                                                                                                   | x       |                   |  |
|     |                                                                                                                                                                   | 0.0     |                   |  |
| **= | It performs exponential (power) calculation on operators and assigns value to the left side operand.<br>Note: x **= y is same as x = x ** y                       | x       |                   |  |
|     |                                                                                                                                                                   | 0.0     |                   |  |
|     |                                                                                                                                                                   | y       |                   |  |
|     |                                                                                                                                                                   | 7       |                   |  |
|     |                                                                                                                                                                   | x **= y |                   |  |
|     |                                                                                                                                                                   | x       |                   |  |
|     |                                                                                                                                                                   | 0.0     |                   |  |
|     |                                                                                                                                                                   | y **= x |                   |  |
|     |                                                                                                                                                                   | y       |                   |  |
|     |                                                                                                                                                                   | 1.0     |                   |  |

## Logical Operators

There are three logical operators supported by Python. These operators are (and, or, not) are to be written in lowercase only. The logical operator evaluates to either True or False based on the logical operands on either side. Every value is logically either True or false. By default values are True except None, False, 0, Empty collection "", {}, {} and few other special values. So, if we say num1 = 10, num2 = -20, then both num1 and num2 are logically True.

### Logical Operators in Python

| Operator | Operation   | Description                                                      | Example             |
|----------|-------------|------------------------------------------------------------------|---------------------|
| and      | Logical AND | If both the operands are True then condition becomes True.       | True and True       |
|          |             |                                                                  | True                |
|          |             |                                                                  | True and False      |
|          |             |                                                                  | False               |
|          |             |                                                                  | False and True      |
|          |             |                                                                  | False               |
|          |             |                                                                  | False and False     |
|          |             |                                                                  | False               |
|          |             |                                                                  | num1                |
|          |             |                                                                  | 10                  |
|          |             |                                                                  | num2                |
|          |             |                                                                  | -20                 |
|          |             |                                                                  | bool(num1 and num2) |
|          |             |                                                                  | True                |
|          |             |                                                                  | num3 = 0            |
| or       | Logical OR  | If any of the two operands are True then condition becomes True. | num1                |
|          |             |                                                                  | 10                  |
|          |             |                                                                  | num2                |
|          |             |                                                                  | -20                 |
|          |             |                                                                  | num3                |
|          |             |                                                                  | 0                   |
|          |             |                                                                  | bool(num1 and num3) |
|          |             |                                                                  | False               |
|          |             |                                                                  |                     |
|          |             |                                                                  | True or True        |
|          |             |                                                                  | True                |
|          |             |                                                                  | True or False       |
|          |             |                                                                  | True                |
|          |             |                                                                  | False or True       |
|          |             |                                                                  | True                |
| not      | Logical NOT | Used to reverse the logical state of its operand.                | False or False      |
|          |             |                                                                  | False               |
|          |             |                                                                  | num1                |
|          |             |                                                                  | 10                  |
|          |             |                                                                  | num2                |
|          |             |                                                                  | -20                 |
|          |             |                                                                  | num3                |
|          |             |                                                                  | 0                   |
|          |             |                                                                  | bool(num1 or num3)  |
|          |             |                                                                  | True                |
|          |             |                                                                  |                     |
|          |             |                                                                  | num1                |
|          |             |                                                                  | 10                  |
|          |             |                                                                  | bool(num1)          |
|          |             |                                                                  | True                |
|          |             |                                                                  | bool(num2)          |
|          |             |                                                                  | True                |
|          |             |                                                                  | bool(num3)          |
|          |             |                                                                  | False               |
|          |             |                                                                  | num3                |
|          |             |                                                                  | 0                   |

```
not num1
False
not num3
True
```

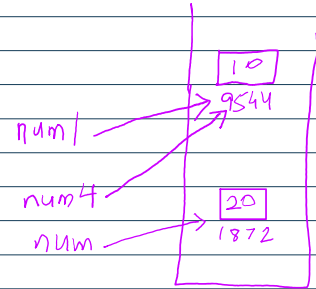
## Identity Operators

Identity operators are used to determine whether the value of a variable is of a certain type or not. Identity operator can also be used to determine whether two variables are referring to the same object or not. There are two identity operators.

### Identity Operators in Python

| Operators | Description                                                                                                                                                                                       | Example                                                                                                    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| is        | Evaluates to True if the variable on either side of the operator points towards the same memory location, and false otherwise. var1 is var2 results to True if id(var1) is equal to id(var2).     | <pre>num1 10 num4 = num1 num4 10 id(num4) 140711737379544 id(num1) 140711737379544 num1 is num4 True</pre> |
| is not    | Evaluates to False if the variable on either side of the operator points towards the same memory location, and True otherwise. var1 is var2 results to True if id(var1) is not equal to id(var2). | <pre>num1 is not num4 False num1 is not num2 True num2 -20 id(num2) 2515072441872</pre>                    |

memory reference →



Both pointing to same memory location.

**Membership operator:** Membership operators are used to check if a value is a member of a given sequence or not.

### Membership operators in Python

| Operator | Description                                                                                        | Example                                                                           |
|----------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| in       | Returns True if the variable or value is found in the specified sequence, and False otherwise.     | <pre>a = [1,2,3,4,5] type(a) &lt;class 'list'&gt; 3 in a True -3 in a False</pre> |
| not in   | Returns True if the variable or value is not found in the specified sequence, and False otherwise. | <pre>-3 not in a True a [1, 2, 3, 4, 5] 3 not in a False</pre>                    |

Sequence (List)

## Expressions

An expression is defined as a combination of constants, variables, and operators. An expression

always evaluates to a value. A value or a standalone variable is also considered as an expression, but a standalone operator is not an expression.

Examples: 100, 3.0 + 3.14, num, num -20.4, 23/5 - 5 \*7 \*(14 -2), "Global" + Citizen"

### Precedence of Operators

Evaluation of the expression is based on precedence of operators. When an expression contains different kind of operators, precedence determines which operator should be applied first. Higher precedence operator is evaluated before the lower precedence operator. Most of the operators studied till now are binary operators. Binary operators are the operators with two operands. The unary operators need only one operand, and they have a higher precedence than the binary operators. The minus as well as plus operators can act as both unary and binary operators. But 'not' is a unary logical operator.

Here's a concise table of operator precedence in Python, listed from highest to lowest precedence:

| Operator                                       | Description                                      |
|------------------------------------------------|--------------------------------------------------|
| ()                                             | Parentheses (used for grouping)                  |
| **                                             | Exponentiation                                   |
| +x, -x, ~x                                     | Unary plus, minus, and bitwise NOT               |
| *, /, //, %                                    | Multiplication, division, floor division, modulo |
| +, -                                           | Addition, subtraction                            |
| <<, >>                                         | Bitwise shift operators                          |
| &                                              | Bitwise AND                                      |
| ^                                              | Bitwise XOR                                      |
| ~                                              | Bitwise NOT                                      |
| (==, !=, >, <, >=, <=), is, is not, in, not in | Comparison, identity, and membership operators   |
| not                                            | Logical NOT                                      |
| and                                            | Logical AND                                      |
| or                                             | Logical OR                                       |
| if ... Else                                    | Conditional expression (ternary operator)        |
| =, +=, -=, *=, /=, %=, //= **=etc.)            | Assignment operators                             |

#### Notes:

Operators higher in the table have higher precedence and are evaluated first.

Operators with the same precedence are evaluated left-to-right, except for \*\* (right-to-left).

Example:  $20 + 30 * 40 = 20 + (30 * 40) = 1,220$

# Precedence of \* is more than that of +.

```
15.0 / 4 + (8.0 + 3.0)
= 15.0 / 4 + 11.0
= 3.75 + 11.0
= 14.75
```

**Statement:** In Python, a statement is a unit of code that the Python interpreter can execute.

```
x = 4           #assignment statement
cube = x ** 3   #assignment statement
print(x, cube)  #print statement
```

4 64

### Input and Output

Sometimes a program needs to interact with the users to get some input data or information from the



end user and process it to give the desired output. In Python we have the input() function. For taking the user input. The input() Function prompts the user to enter data. It accepts all user input as a string. The user may enter a number or string, but the input() function treats them as a string only. The syntax for input() is:

```
input([Prompt])
```

'Prompt' Prompt is a string we may like to display on the screen prior to taking the input and it is optional. When a prompt is specified first, it is displayed on the screen, after which the user can enter data. The input () takes exactly what is typed from the keyboard and converts it into a string and assigns it to a variable on the left hand side of the assignment operator(=). Entering data for the input function is terminated by pressing the Enter key.

```
fname = input('Enter your first name: ')
age = input('Enter your age: ')
tp = type(age)
print(tp)
```

Output:

```
Enter your first name: Vaanya
Enter your age: 11
<class 'str'>
```

```
fname = input('Enter your first name: ')
age = int(input('Enter your age: '))
tp = type(age)
print(tp)
```

Output:

```
Enter your first name: Vaanya
Enter your age: 11
<class 'int'>
```

Python uses the print() function to output the data to standard output device, the screen or terminal. The function print() evaluates the expression before displaying it on the screen. The print() Outputs a complete line and then move moves to the next line for a subsequent output. The syntax for print () is:

```
print(value1, value2, ... [,..., sep = ' ', end = '\n'])
```

- **sep:** The optional parameter Sep is a separator between the output values. We can use a character, integer or a string as a separator. The default separator is space.
- **end:** This is also optional and it allows u.... last value. The default value is a new line character.

## Type Conversion

Consider a code:

```
Num1 = input('Enter a number and I'll double it:')
Num1 = Num1 * 2
print(Num1)
```

Output:

```
Enter a number and I'll double it: 3
33
```

To get as output, we need to convert the data type of the value entered by the user to integer. Thus, we modify the program as follows:

```
num1 = input("Enter a number and I'll double it: ")
num1 = int(num1) * 2
print(num1)
```

Output:

```
Enter a number and I'll double it: 5
10
```

As and when required, we can change the data type of the variable in Python from 1 type to another. Such data type conversion can happen in two ways: either explicitly (forced) when the programmer specifies for the interpreter to convert a data type to another type. Or implicitly when the interpreter understand such a need by itself and does the type conversion automatically.

### Explicit Conversion:

**Explicit conversion, also called typecasting, happens when the data type conversion takes place because the programmer forced it in the Program.** The general form of an explicit data type conversion is:

(new data\_type)(expression)

With explicit type conversion, there is a risk of loss of information. Since we are forcing an expression to be of a specific type. For example, converting a float value  $X = 20.67$ . An integer type that is `int(X)` Will discard the fractional part  $0.67$ . Following are some of the functions in Python that are used for explicit converting an expression or a variable to a different type.

| Function               | Description                            |
|------------------------|----------------------------------------|
| <code>int(x)</code>    | Converts x to an integer.              |
| <code>float(x)</code>  | Converts x to a floating point number. |
| <code>str(x)</code>    | Converts x to a string representation. |
| <code>chr(x)</code>    | Converts x to a character.             |
| <code>unichr(x)</code> | Converts x to a Unicode character.     |

# Explicit type conversion from int to float.

```
num1 = 10
num2 = 20
num3 = num1 + num2
print(num3)
print(type(num3))
num4 = float(num1 + num2)
print(num4)
print(type(num4))
```

Output:

```
30
<class 'int'>
30.0
<class 'float'>
```

# Explicit type conversion from float to int

```
num1 = 10.2
num2 = 20.6
num3 = num1 + num2
print(num3)
print(type(num3))
num4 = int(num1 + num2)
```

```

print(num3)
print(type(num3))
num4 = int(num1 + num2)
print(num4)
print(type(num4))

```

Output:

```

30.8
<class 'float'>
30
<class 'int'>

```

```

priceIcecream = 25
priceBrownie = 45
totalPrice = priceIcecream + priceBrownie
print('The total is ₹:', totalPrice)

```

Output:

```
The total is ₹: 70
```

```

# Conversion between numbers and strings.
priceIcecream = 25
priceBrownie = 45
totalPrice = priceIcecream + priceBrownie
print('The total is ₹:' + totalPrice)

```

Trying to concatenate int to str

Output:

```

print('The total is ₹:' + totalPrice)
~~~~~^~~~~~
TypeError: can only concatenate str (not "int") to str

```

```

# Conversion between numbers and strings.
priceIcecream = 25
priceBrownie = 45
totalPrice = priceIcecream + priceBrownie
print('The total is ₹:' + str(totalPrice))

```

↑ explicit type conversion.

Output:

```
The total is ₹:70
```

WAPP to input the price of ice-cream, brownie, coke.  
Add them up and print the total cost on the terminal.

### Implicit Conversion

Implicit conversion, also known as coercion, happens when data type conversion is done automatically by Python and is not instructed by the programmer.

```
# Implicit type conversion from int to float.
num1 = 10
num2 = 20.0
num3 = num1 + num2 # Num3 is sum of an int and float.
print(num3)
print(type(num3))
```

Output:

```
30.0
<class 'float'>
```

In the above example, an integer value stored in variable num1 is added to the float value stored in variable num2, and the result was automatically converted to a float value stored in a variable num3 without explicitly telling the interpreter. This is an example of implicit data conversion. One may wonder why was the float value not converted to an integer instead? This is due to type promotion that allows performing operations (whenever possible) By converting data into a wider-sized data type without any loss of information.

## Debugging

A programmer can make mistakes while writing a program and hence the program may not execute or may generate wrong output. The process of identifying and removing such mistakes, also known as bugs or errors, from a program, is called debugging. Errors occurring in program can be categorized as:

- i) Syntax Error: Like other programming languages, Python has its own rules that determine its syntax. The interpreter interprets the statement only if it is syntactically correct. If any syntax error is present, the interpreter shows error messages and stops the execution there.
- ii) Logical Error: A logical error is a bug in the program that causes it to behave incorrectly. A logical error produces an undesired output, but with abrupt termination of the execution of the program. Since the program interprets successfully even when logical errors are present in it, it is sometimes difficult to identify these errors. The only evidence to the existence of logical error is the wrong output. While working backwards from the output of the program 1 can identify what went wrong.
- iii) Runtime Errors: A runtime error causes abnormal termination of a program while it is executing. Runtime error is when the statement is correct syntactically but the interpreter cannot execute it. Runtime errors do not appear until after the program starts running or executing. For example, we have a statement having division operations in the program. By mistake if the denominator entered is zero then it will give a runtime error like division 0.

```
number = int(input('Enter numerator: '))
deno = int(input('Enter denominator: '))
div = number / deno
print('Result: ', div)
```

Output:

```
Enter numerator: 200
Enter denominator: 0
Traceback (most recent call last):
  File "c:\Users\bluej\Documents\Students\
e>
  div = number / deno
ZeroDivisionError: division by zero
```